Jack Ziyang Cai

PHONE: (314)-201-1302 | EMAIL: zc5794@princeton.edu

Education

- **Ph.D. In Computer Science** **2025 - Present**
  Princeton University *Princeton, New Jersey*
  Princeton Language and Intelligence Institute

- **Master of Science in Electrical and Computer Engineering** **2023 - 2025**
  University of Wisconsin-Madison *Madison, United States*
  GPA: 3.91

- **Bachelor of Science in Computer Engineering** **2019 - 2023**
  Concurrent major in Computer Science and Math *Madison, United States*
  University of Wisconsin-Madison
  GPA: 3.86, Honors in Major

Publications

Under review

- **AbridgeKV: Compressing Long Reasoning into the KV Cache of Learned Pause Tokens**

  ***Ziyang Cai**, Lijie Yang, Vasilis Kontonis, Shivam Garg*

  Under Review, 2026

- **Fork-and-Flush: Escaping Idea Basins in Autoresearch Agents**

  ***Ziyang Cai**, Christos Ziakas, Vasilis Kontonis, Tim Pearce, Siddhartha Sen, Akshay Krishnamurthy, Shivam Garg, Dimitris Papailiopoulos*

  Under Review, 2026

- **T^{2}MLR: Transformer with Temporal Middle-Layer Recurrence**

  Ziyang Cai, Xingyu Zhu*, Yihe Dong, Yinghui He, Sanjeev Arora*

  Under Review, 2026; LIT Workshop @ ICLR 2026

- **SoftSphere: Understanding and Mitigating Collapse in Latent Reasoning**

  *Yihe Dong, **Ziyang Cai***

  Under Review, 2026

- **Goedel-Architect: Streamlining Formal Theorem Proving with Blueprint Generation and Refinement**

  *Jui-Hui Chung, **Ziyang Cai**, Zihao Li, Qishuo Yin, Rohit Agarwal, Simon Park, Rodrigo Porto, Narutatsu Ri, Ziran Yang, Shange Tang, Xingyu Dang, Hongzhou Lin, Mengdi Wang, Danqi Chen, Chi Jin, Liam H Fowl, Sanjeev Arora*

  Under Review, 2026; AI4Math Workshop @ ICML 2026

- **Full-bandwidth Transformer**

  *Xi Wang, **Ziyang Cai**, Zheng Zhan, Harry Dong, Ying Fan, Gustavo Henrique de Rosa, Tim Pearce, John Langford*

  Under Review, 2026

Published

- **Synthetic Tasks for Training AutoResearch**

  ***Ziyang Cai**, Amir Saeidi, Harkirat Behl*

  NeurIPS, 2026

- **h1: Bootstrapping LLMs to Reason over Longer Horizons via Reinforcement Learning**

  *Alesia Ivanova, Sumeet Ramesh Motwani, **Ziyang Cai**, Philip Torr, Riashat Islam, Shital Shah, Christian Schroeder de Witt, Charles London*

  ICML Spotlight, 2026

- **Extrapolation by Association: Length Generalization Transfer in Transformers**

  ***Ziyang Cai**, Nayoung Lee, Avi Schwarzschild, Samet Oymak, Dimitris Papailiopoulos*

  NeurIPS Spotlight, 2025

- **Self-Improving Transformers Overcome Easy-to-Hard and Length Generalization Challenges**

  Nayoung Lee, **Ziyang Cai***, Avi Schwarzschild, Kangwook Lee, Dimitris Papailiopoulos*

  ICML, 2025; Oral, SSI-FM Workshop @ ICLR 2025

- **Everything Everywhere All at Once: LLMs can In-Context Learn Multiple Tasks in Superposition**

  *Zheyang Xiong, **Ziyang Cai**, John Cooper, Albert Ge, Vasilis Papageorgiou, Zack Sifakis, Angeliki Giannou, Ziqian Lin, Liu Yang, Saurabh Agarwal, Grigorios Chrysos, Samet Oymak, Kangwook Lee, Dimitris Papailiopoulos*

  ICML Spotlight, 2025

- **R&B: Domain Regrouping and Data Mixture Balancing for Efficient Foundation Model Training**

  *Albert Ge, Tzu-Heng Huang, John Cooper, Avi Trost, Ziyi Chu, Satya Sai Srinath Namburi GNVV, **Ziyang Cai**, Kendall Park, Nicholas Roberts, Frederic Sala*

  DataWorld & DIG-BUG Workshops, 2025

- **Delving into Out-of-Distribution Detection with Vision-Language Representations**

  *Yifei Ming, **Ziyang Cai**, Jiuxiang Gu, Yiyou Sun, Wei Li, Yixuan Li*

  NeurIPS, 2022

* Equal contribution

Research Experience

- **PhD Student in Sanjeev Arora's Lab** **Sep. 2025 -Present**

  Princeton University *Princeton, NJ*

  – Focus on language model architectures, AI for Math, Pretraining, Coding & auto-research agents.

  – Interest in synthetic data generation, theory of language models, and out-of-distribution generalization.

- **Research Intern** **May 2026 - Aug. 2026**

  Microsoft Research *New York, NY*

- **Research Intern** **Jun. 2025 - Aug. 2025**

  Microsoft Research *Seattle, WA*

- Created automated synthetic data pipelines for training coding agents.
- Synthesized hundreds of ML coding tasks and thousands of agent trajectories for training.
- Trained 4B and 8B models to perform on-par with frontier models on the MLGym benchmark.

- **Research Assistant** **Aug. 2023 - Aug. 2025**
  Prof. Dimitris Papailiopoulos's Lab *UW-Madison*
  - Investigating in-context learning and out-of-distribution generalization in language models.
  - Demonstrated superposition phenomenon in pretrained language models, analyzing how they solve multiple superimposed ICL tasks through the lens of task vectors.
  - Demonstrated weak-to-strong generalization and length generalization in transformer models trained on synthetic problems.
  - Investigating compositional generalization in transformers on synthetic problems.

- **Research Assistant** **Jul. 2022 - Aug. 2023**
  Prof. Junjie Hu's Lab *UW-Madison*
  - Researched Scene Graph Generation using pretrained multimodal models like BLIP and OFA.
  - Fine-tuned open-source multimodal models on scene graph data, achieving comparable results with other conventional multi-staged methods.

- **Research Assistant** **Sep. 2021 - Jun. 2022**
  Prof. Sharon Li's Lab *UW-Madison*
  - Helped develop a new out-of-distribution detection method in image classification using visual-lingual representations from the CLIP model.
  - Contributed to research demonstrating the superiority of pretrained vision-language models in OOD detection compared to traditional approaches.

Professional Service

- **Reviewer:** ICLR 2027, ICML 2026, COLM 2024
- **Program Committee:** AAAI 2027

Work Experience

- **Project and Program Management Researcher** **Oct. 2025 - Dec. 2025 (Part-time)**
  Aquent LLC (placed with Microsoft AI Frontiers Lab) *Remote / Princeton, NJ*
- **Teaching Assistant** **Spring 2024**
  ECE 431 Digital Signal Processing *UW-Madison*
- **Teaching Assistant** **Fall 2023**
  ECE 561 Information Theory and Machine Learning *UW-Madison*
- **Software Engineer Intern** **May 2022 - Aug. 2022**
  Cisco Systems Inc. *San Jose, CA*

Projects

- **CAFA 5 Protein Function Prediction Challenge (Kaggle)** **2023**
  - Developed a deep learning pipeline to predict hierarchical labels for protein function.
  - Leveraged diverse data sources including protein sequences, AlphaFold protein structures, and protein-protein interaction graphs.
  - Implemented a graph neural network and protein language models to generate features for prediction.

Technical Skills

- **Programming Languages:** Python, C++
- **Tools and Frameworks:** PyTorch, TensorFlow, Hugging Face, CUDA
- **Areas of Expertise:** Machine Learning, Natural Language Processing, Computer Vision, Out-of-Distribution Detection, In-Context Learning